COMPUTER NETWORKS (MIC23) — UNIT II

Data Link Layer & Error Control — Handwritten Study Notes

Student Blue-Pen Notebook Style • Short & Direct 2M Answers • Complete Visual Diagrams for Every Question • All Source Papers

This Master Question & Answer document combines all exam questions from:

- **Model Paper-1** (MIC23 B.Tech CSE/AIML)
- **Model Paper-2** (MIC23 B.Tech CSE/AIML)
- **Previous University Exam Paper** (April-2026 Regular/Supple)
- **Mid-1 Important Questions**

All answers are rewritten in **extremely simple English** that any beginner student can read once and understand immediately. Every question includes a relevant visual diagram, real-life analogies, box comparison tables, step-by-step numerical solutions, and full examination structures.

SECTION A: 2-MARK QUESTIONS & ANSWERS

Short, direct answers (~40–65 words) written in simple English with student pen-style diagrams

(Short, direct answers written in simple English with small visual diagrams)

Q1. *(Paper-1)*

Enumerate any four responsibilities of the Data Link Layer in OSI Model?

1. Meaning in Simple English

The **Data Link Layer** is Layer 2 of the OSI model. Its job is to take raw data packets from the Network Layer and deliver them safely across a single physical wire to the next computer.

2. Four Main Responsibilities

- **Framing:** Divides the stream of bits received from Network Layer into distinct, manageable blocks called **Frames**.
- **Physical (MAC) Addressing:** Adds the sender's and receiver's physical hardware address (MAC Address) to each frame.
- **Flow Control:** Prevents a super-fast sending computer from drowning a slow receiving computer with too much data.
- **Error Control:** Detects damaged, lost, or duplicate frames using checksums/CRC and retransmits them safely.

3. Diagram Explanation

The diagram below shows the Data Link Layer packaging network packets into frames, adding Header (MAC addresses) and Trailer (CRC error check), and sending them over the physical wire.

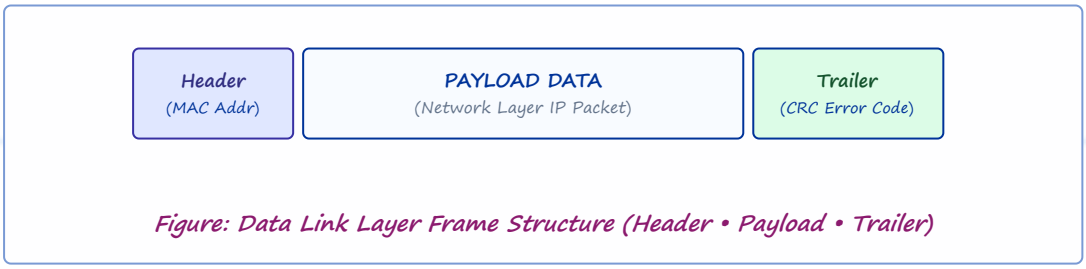


Figure: Data Link Layer Frame Structure (Header • Payload • Trailer)

Q2. *(Paper-1)*

What are the different types of Errors that occur in the data in a network, give Example?

1. Meaning in Simple English

When electrical or radio signals travel across cables, noise, lightning, and interference can flip 0s into 1s, or 1s into 0s. This corruption is called an **Error**.

2. Two Types of Errors

Error Type	Definition in Simple English	Clear Example
Single-Bit Error	Only **one single bit** in the entire data block gets flipped from 0 to 1 or 1 to 0.	Sent: 00000000 Received: 00000010 (Only 1 bit changed).
Burst Error	**Two or more consecutive bits** get corrupted by a sudden burst of electrical noise.	Sent: 01000100 Received: 01111100 (Several bits flipped in a row).

3. Diagram Explanation

The diagram shows a single corrupted bit versus a cluster of corrupted bits caused by noise.

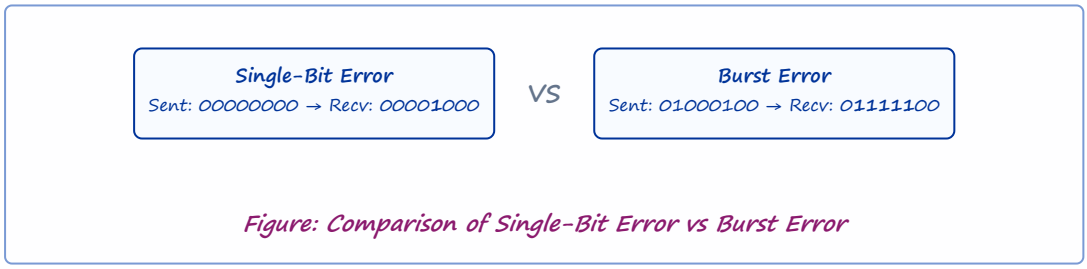


Figure: Comparison of Single-Bit Error vs Burst Error

Q3. *(Paper-2)*

Explain the importance of framing in the Data Link Layer?

1. Meaning in Simple English

The physical layer just transmits an endless, continuous stream of raw electrical pulses: `01011100010101...` Without boundaries, the receiver has no idea where one message ends and the next message begins.

2. Why Framing is Essential

- **Defines Boundaries:** Separates the continuous stream of bits into distinct, readable units called **Frames** by adding clear start and end flags.
- **Carries Delivery Info:** Attaches a **Header** containing source and destination MAC hardware addresses.
- **Carries Error Checking:** Attaches a **Trailer** with CRC error-detection codes so the receiver can verify if the frame arrived cleanly.

3. Diagram Explanation

The diagram shows a raw stream of bits being converted into discrete frames separated by start and end delimiter flags.



Figure: Framing Boundaries Defined by Start and End Delimiter Flags

Q4. *(Paper-2)*

Define flow control and explain why it is essential?

1. Meaning in Simple English

Flow Control is a set of rules that tells the sender computer how fast it is allowed to send data.

2. Why It is Essential

- **Protects Receiver Buffer:** If a powerful server sends data at 100 Mbps, but an old smartphone receiver can only process data at 10 Mbps, the phone's memory buffer will overflow and drop packets.
- **Eliminates Packet Loss:** Flow control forces the fast sender to slow down and wait until the receiver has finished processing previous frames.

3. Diagram Explanation

The diagram shows a fast sender paused by an "ACK / Buffer Full" feedback signal from a slow receiver.

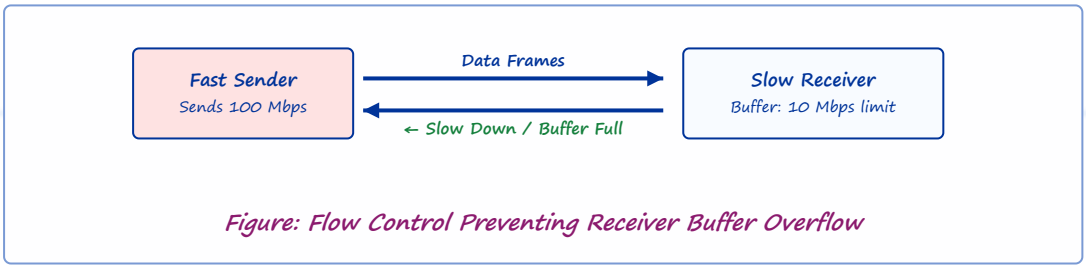


Figure: Flow Control Preventing Receiver Buffer Overflow

Q5. *(Paper-3)*

What is one's complement checksum? Where is it used?

1. Meaning in Simple English

One's complement checksum is an error-detection technique. The sender splits data into equal 8-bit or 16-bit binary words, adds them together using one's complement arithmetic (carrying extra overflow bits back to the bottom), and then flips every bit (0 becomes 1, and 1 becomes 0).

2. Where It Is Used

- Extensively used in **TCP/IP Internet protocols**: **IPv4 Header Checksum**, **TCP Segment Checksum**, and **UDP Checksum**.
- Used to ensure headers have not been corrupted during router hops.

3. Diagram Explanation

The diagram shows data words added together, extra carries folded back, and the sum inverted to produce the final checksum.



Figure: One's Complement Checksum Generation Steps

Q6. *(MID-1)*

What is CRC (Cyclic Redundancy Check)?

1. Meaning in Simple English

CRC (Cyclic Redundancy Check) is a very powerful, mathematical error-detection technique based on binary polynomial division.

2. How It Works

- The sender appends k zero bits to the data (where k is the degree of a predetermined generator polynomial).
- The sender divides this number by the generator binary divisor using **modulo-2 division (XOR)**.
- The resulting remainder is called the **CRC Checksum**.
- The sender attaches this remainder to the data and transmits it. If the receiver divides the incoming frame by the same divisor and gets a remainder of zero, there is no error.

3. Diagram Explanation

The diagram shows Data + Zeros divided by a Divisor to produce a CRC Remainder, which replaces the zeros for transmission.

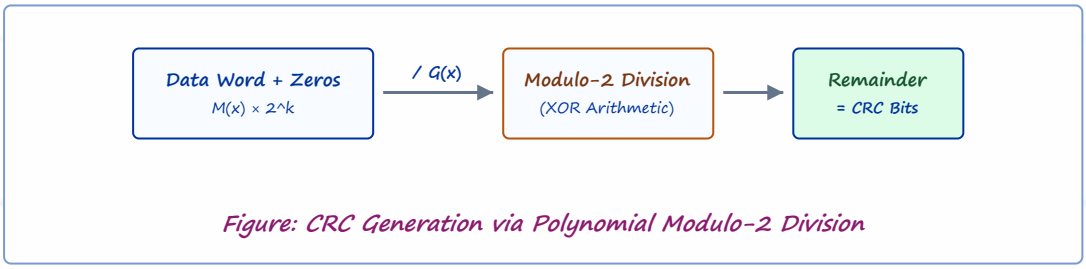


Figure: CRC Generation via Polynomial Modulo-2 Division

Q7. *(MID-1)*

Explain the basic idea of the Stop-and-Wait protocol.

1. Meaning in Simple English

The **Stop-and-Wait protocol** is the simplest flow and error control protocol. The sender transmits **one single frame** and then completely **stops and waits**.

2. Core Working Rules

- The sender starts a timer and waits for the receiver to send an **Acknowledgment (ACK)**.
- Only after receiving the ACK does the sender send the next frame.
- If the timer runs out before the ACK arrives (timeout), the sender assumes the frame was lost and retransmits it.

3. Diagram Explanation

The diagram shows Sender sending Frame 0, waiting idle, receiving ACK 1, and only then sending Frame 1.

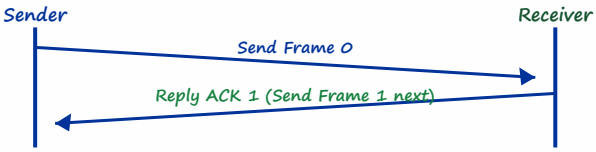


Figure: Stop-and-Wait Alternating Transmission Cycle

SECTION B: 5-MARK QUESTIONS (FULL 10-MARK EXAM STRUCTURE)

Detailed answers with Introduction, Dual Diagrams (Standard Exam + Intuitive Analogy), Box Comparison Tables, and Step-by-Step Numerical Calculations

(All prepared using complete 10-Mark Answer Structure: Introduction, Definition, Detailed Points, Dual Diagrams, Step-by-Step Calculations, Real-Life Examples, Box Tables, and Conclusions)

Q1. *(Paper-1)*

What is Cyclic Redundancy Check? Calculate CRC code using CRC12 for the following data: 100100, by using the following generator polynomial: x^{12} + x^{11} + x^3 + x^2 + x + 1?

1. Introduction

In computer networks, electrical noise on transmission lines can flip bits. Simple parity checks can only detect a single flipped bit. To detect burst errors reliably, network engineers use **CRC (Cyclic Redundancy Check)**, which is used in Ethernet, Wi-Fi, and storage hard drives.

2. Definition

Cyclic Redundancy Check (CRC) is a polynomial-based block error detection code. It operates by dividing a binary data block by a predefined generator polynomial using modulo-2 (XOR) binary arithmetic, appending the remainder to the message as redundant check bits.

3. Step-by-Step Numerical Problem Solution

1. Given Data

- **Data Word:** M = 100100 (6 bits).
- **Generator Polynomial:** G(x) = x^{12} + x^{11} + x^3 + x^2 + x + 1 (CRC-12).

2. What We Need to Find

- Generator Divisor Bit String.
- Number of zero bits to append (k).
- Remainder generated using Modulo-2 (XOR) binary division.
- Final CRC Codeword to be transmitted.

3. Binary Conversion of the Polynomial

- We write the coefficient (1 or 0) for every power of x from x^{12} down to x^0:
- x^{12} = 1
- x^{11} = 1
- x^{10} = 0
- x^9 = 0
- x^8 = 0
- x^7 = 0
- x^6 = 0
- x^5 = 0
- x^4 = 0
- x^3 = 1
- x^2 = 1
- x^1 = 1
- x^0 = 1

- **Divisor Bit String:** 1 1 0 0 0 0 0 0 0 1 1 1 1 (13 bits long).
- Degree of the polynomial is **k = 12**.

4. Preparing the Dividend

- Append k = 12 zeros to the data word (100100):
- **Dividend:** 1 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 (Total 6 + 12 = 18 bits).

5. Modulo-2 (XOR) Long Division

In Modulo-2 arithmetic, subtraction is identical to **XOR**:

- $0 \oplus 0 = 0$
- $1 \oplus 1 = 0$
- $1 \oplus 0 = 1$
- $0 \oplus 1 = 1$

Let us perform the division:

Divisor: 1100000001111 (13 bits)

Dividend: 100100000000000000

Step 1:

1001000000000 (first 13 bits)

^ 1100000001111 (Divisor)

= 0101000001111

Bring down next bit '0' -> 1010000011110 (13 bits, starts with 1 -> XOR)

^ 1100000001111

= 0110000010001

Bring down next bit '0' -> 1100000100010 (13 bits, starts with 1 -> XOR)

^ 1100000001111

= 0000000101101

Bring down next bit '0' -> 0000001011010 (Starts with 0 -> XOR with 0000000000000)

Bring down next bit '0' -> 0000010110100

Bring down next bit '0' -> 0000101101000

Final 12-bit Remainder (CRC bits) = 000101101000

6. Form the Final Codeword

- ext{Transmitted Codeword} = ext{Data Word} + ext{Remainder}
- ext{Codeword} = 100100\ 000101101000

7. Final Boxed Answer

CRC-12 Remainder = 000101101000

Final Transmitted Codeword = 100100000101101000

8. Verification at Receiver

When the receiver receives 100100000101101000, it divides it by 1100000001111 using XOR. The remainder will be **000000000000**, proving 100% error-free transmission.

4. Real-Life Analogy

CRC is like the **Checksum on a Credit Card or Bank Note**:

- The bank prints 16 digits on your card, plus 1 security digit calculated by a mathematical formula.
- When a shop swipes your card, the terminal checks the math. If a digit was entered incorrectly, the math fails immediately and rejects the transaction.

5. Dual Diagrams

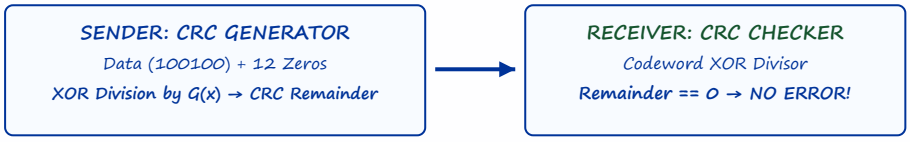


Diagram 1 (Standard Exam Diagram): CRC-12 Generation and Receiver Zero-Remainder Verification



Diagram 2 (Easy Understanding Diagram): Credit Card Check Digit Analogy for CRC

Diagram 1: Standard Exam Diagram

Shows the Sender side CRC generation block (Data + Zeros o Modulo-2 Division o CRC Remainder added o Transmitted Codeword).

Diagram 2: Easy Understanding Diagram

Shows the Receiver verification: Incoming Codeword divided by Divisor. Remainder = 0 means Accept; Remainder

eq 0 means Discard.

6. Advantages and Disadvantages of CRC

Advantages	Disadvantages
Detects 100% of Single-Bit Errors.	Slightly more complex mathematical computation.

Advantages	Disadvantages
Detects 100% of all Burst Errors shorter than or equal to polynomial degree k.	Adds extra bits (k bits) to every transmitted frame.
Extremely Fast in Hardware: Can be implemented cheaply using simple shift registers and XOR gates.	Does not automatically correct errors; only detects them.

7. Conclusion

CRC is the standard error-checking method used in Ethernet and Wi-Fi because it catches almost all burst errors very quickly without slowing down the computer.

Q2. *(Paper-1)*

Explain Stop and Wait ARQ protocol with diagram?

1. Introduction

Simple Stop-and-Wait assumes that transmission channels are perfect and no packets are ever lost. However, real cables experience noise, packet drops, and hardware faults. To make communication reliable over noisy channels, engineers invented **Stop-and-Wait ARQ (Automatic Repeat reQuest)**.

2. Definition

Stop-and-Wait ARQ is a flow and error control protocol where the sender transmits a single frame, starts a retransmission timer, and waits for a positive acknowledgment (ACK). If the ACK does not arrive before the timer expires, the sender automatically retransmits the frame.

3. Three Key Mechanisms Used in Stop-and-Wait ARQ

- **Sequence Numbers (0 and 1):** Frames are numbered alternately as Frame 0 and Frame 1. This prevents the receiver from accepting duplicate copies if an ACK was delayed.
- **Retransmission Timer (Time-out):** When the sender transmits a frame, it starts a countdown clock. If no ACK arrives before the clock reaches zero, it resends the same frame.
- **Acknowledgment Numbers (ACK 0 and ACK 1):** The receiver sends an ACK indicating the sequence number of the next frame it expects to receive.

4. Three Operational Scenarios

Scenario A: Normal Flawless Operation

- Sender sends **Frame 0**.

- Receiver receives Frame 0 safely and sends **ACK 1** (meaning: *"I got Frame 0, send Frame 1 next"*).
- Sender receives ACK 1 and transmits **Frame 1**.

Scenario B: Frame Lost in Transit

- Sender sends **Frame 0**, but electrical noise destroys it before it reaches the receiver.
- The receiver never gets it, so it sends nothing.
- The sender's **Time-out Timer expires**.
- Sender retransmits **Frame 0**. Receiver receives it and replies with **ACK 1**.

Scenario C: Acknowledgment (ACK) Lost

- Sender sends **Frame 0**. Receiver gets it and sends **ACK 1**.
- The ACK 1 packet gets lost on the way back.
- The sender's timer expires and it retransmits **Frame 0**.
- The receiver sees Frame 0 again, recognizes it as a **duplicate**, discards the duplicate copy, and resends **ACK 1**.

5. Real-Life Analogy

Imagine a walkie-talkie conversation:

- You say: *"Team Alpha, we have arrived at the base. Over."* (Frame 0).
- You wait quietly. You do not say anything else.
- The captain replies: *"Copy that, Team Alpha. What is your status? Over."* (ACK 1).
- If you hear only static for 30 seconds, you repeat: *"Team Alpha, we have arrived at the base. Over."* (Retransmission).

6. Dual Diagrams

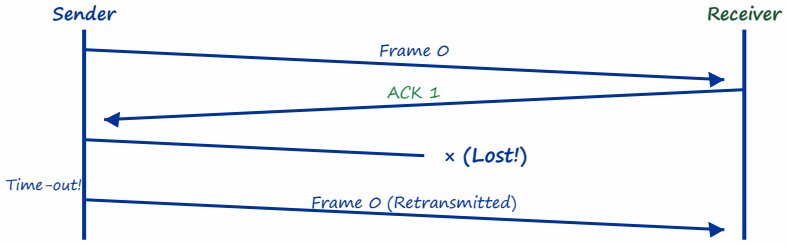


Diagram 1 (Standard Exam Diagram): Stop-and-Wait ARQ Normal Transmission & Lost Frame Retransmission



Diagram 2 (Easy Understanding Diagram): Walkie-Talkie Conversation Analogy for Stop-and-Wait

Diagram 1: Standard Exam Diagram

Shows vertical timelines for Sender and Receiver illustrating the 3 scenarios: Normal Operation, Lost Frame with Timer Expiration, and Lost ACK with Duplicate Discard.

Diagram 2: Easy Understanding Diagram

Shows the Sender with a stopwatch timer ticking down, waiting for a green ACK envelope.

7. Comparison: Stop-and-Wait vs Stop-and-Wait ARQ

Feature	Simple Stop-and-Wait	Stop-and-Wait ARQ
Channel Type	Assumes noiseless, perfect channel.	Handles noisy, error-prone channels.
Sequence Numbers	No sequence numbers needed.	Uses 1-bit sequence numbers (0 and 1).
Timers	No timers.	Uses Retransmission Timer (Time-out).
Lost Packet Handling	Sender freezes forever (deadlock).	Automatically retransmits lost packets.

8. Advantages and Disadvantages

Advantages	Disadvantages
Extremely Simple: Easy to implement in software and hardware.	**Extremely Low Efficiency:** The sender spends most of its time sitting idle waiting for ACKs.
No Large Buffers Needed: Sender only stores 1 frame; receiver only holds 1 frame.	**Wastes Bandwidth:** On long-distance links (satellite), channel utilization drops below 1%.

9. Conclusion

Stop-and-Wait ARQ provides reliable, error-free communication with minimal memory requirements, making it ideal for simple low-speed wireless sensors, but inefficient for high-speed fiber networks.

Q3. *(Paper-1)*

Explain how Hamming code can be used for Error detection in computer networks? Calculate 7 bit Hamming code for the following data: 1011?

1. Introduction

Most error-control codes only detect errors and require the sender to retransmit the data. However, in space probes or real-time streaming, asking for a retransmission takes too much time. In 1950, Richard Hamming invented the **Hamming Code**, which not only detects where an error occurred, but **automatically corrects** the corrupted bit on the spot.

2. Definition

Hamming Code is a forward error-correcting (FEC) code that adds multiple redundant parity bits at specific power-of-two bit positions (1, 2, 4, 8, \dots). The receiver evaluates these parity checks to determine the exact position of any single-bit error and flips it back to its correct value.

3. Step-by-Step Numerical Problem Solution

1. Given Data

- **Data bits:** $d = 1011$ (4 bits: $d_7\ d_6\ d_5\ d_3$).
- **Total Codeword Length:** 7 bits ($n = 7$).
- Parity type: **Even Parity**.

2. What We Need to Find

- Redundant parity bits p_1, p_2, p_4.
- The 7-bit transmitted Hamming Codeword ($b_7\ b_6\ b_5\ b_4\ b_3\ b_2\ b_1$).

3. Bit Position Table

Parity bits are placed at positions that are powers of 2 ($2^0=1, 2^1=2, 2^2=4$):

- Bit 1: p_1 (Parity bit)
- Bit 2: p_2 (Parity bit)
- Bit 3: $d_3 = 1$ (Data bit)
- Bit 4: p_4 (Parity bit)
- Bit 5: $d_5 = 0$ (Data bit)
- Bit 6: $d_6 = 1$ (Data bit)
- Bit 7: $d_7 = 1$ (Data bit)

4. Formula & Parity Calculation (Even Parity)

- **Parity Bit p_1:** Checks positions with a 1 in least significant bit of binary representation (Positions 1, 3, 5, 7):

$p_1 \oplus d_3 \oplus d_5 \oplus d_7 = 0$

$p_1 \oplus 1 \oplus 0 \oplus 1 = 0 \Rightarrow p_1 \oplus 0 = 0 \Rightarrow \mathbf{p_1 = 0}$

- **Parity Bit p_2:** Checks positions with a 1 in second binary digit (Positions 2, 3, 6, 7):

$p_2 \oplus d_3 \oplus d_6 \oplus d_7 = 0$

$p_2 \oplus 1 \oplus 1 \oplus 1 = 0 \Rightarrow p_2 \oplus 1 = 0 \Rightarrow \mathbf{p_2 = 1}$

- **Parity Bit p_4:** Checks positions with a 1 in third binary digit (Positions 4, 5, 6, 7):

$p_4 \oplus d_5 \oplus d_6 \oplus d_7 = 0$

$p_4 \oplus 0 \oplus 1 \oplus 1 = 0 \Rightarrow p_4 \oplus 0 = 0 \Rightarrow \mathbf{p_4 = 0}$

5. Assemble the 7-Bit Codeword

Placing calculated bits into their slots:

- $b_7 = 1$
- $b_6 = 1$
- $b_5 = 0$
- $b_4 = p_4 = 0$
- $b_3 = 1$
- $b_2 = p_2 = 1$
- $b_1 = p_1 = 0$

6. Final Boxed Answer

Calculated Parity Bits: $p_1 = 0,\ p_2 = 1,\ p_4 = 0$

Final 7-Bit Transmitted Hamming Codeword = 1010110

7. How the Receiver Detects & Corrects an Error

Suppose Bit 6 flips during transmission, so receiver gets `1010110` with bit 6 flipped to `0`: `1000110`.

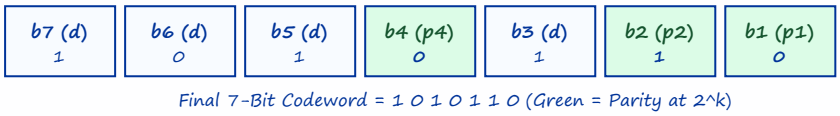
- Check $C_1 = b_1 \oplus b_3 \oplus b_5 \oplus b_7 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$
- Check $C_2 = b_2 \oplus b_3 \oplus b_6 \oplus b_7 = 1 \oplus 1 \oplus 0 \oplus 1 = 1$
- Check $C_4 = b_4 \oplus b_5 \oplus b_6 \oplus b_7 = 0 \oplus 0 \oplus 0 \oplus 1 = 1$
- Syndrome word: $C_4\ C_2\ C_1 = 110_2 = \mathbf{6}$ in decimal!
- The receiver immediately knows **Bit 6 is corrupt**, flips it from `0` back to `1`, and recovers the perfect original data!

4. Real-Life Analogy

Hamming code is like a **GPS Triangulation System**:

- Three satellites check your position from three different angles.
- If you move off-road, the intersection of the three signals pinpoints your exact location immediately.

5. Dual Diagrams



Final 7-Bit Codeword = 1 0 1 0 1 1 0 (Green = Parity at 2^k)

Diagram 1 (Standard Exam Diagram): 7-Bit Hamming Codeword Bit Position Mapping



Diagram 2 (Easy Understanding Diagram): 3-Way Triangulation Pinpoints Corrupted Bit Position

Diagram 1: Standard Exam Diagram

Shows the 7-bit slot table with power-of-two parity positions (p_1, p_2, p_4) and data positions (d_3, d_5, d_6, d_7).

Diagram 2: Easy Understanding Diagram

Shows 3 overlapping Venn diagram circles checking parity, where the intersecting region identifies the exact corrupted bit.

6. Advantages and Disadvantages of Hamming Code

Advantages	Disadvantages
Instant Error Correction: No need to retransmit; corrects errors immediately.	Adds many extra bits (3 check bits for every 4 data bits = 43% extra data).
Ideal for Long Delays: Perfect for satellite links and ECC computer RAM.	Can only reliably correct **single-bit errors**.

7. Conclusion

Hamming code is a masterpiece of mathematical engineering. It turns passive error detection into active error correction, ensuring reliable performance in space communications and server memory.

Q4. *(Paper-1)*

Explain the services provided by Data link layer?

1. Introduction

The **Data Link Layer (DLL)** is Layer 2 of the OSI Reference Model. It sits directly between the Physical Layer below and the Network Layer above, turning the unreliable raw bit-pipe of cables into a smooth, reliable communication channel.

2. Definition

The Data Link Layer is responsible for node-to-node (hop-to-hop) frame transmission, physical MAC addressing, flow control, error detection and recovery, and medium access regulation over a single physical link.

3. Core Services Provided by Data Link Layer

A. Framing

- Packages network layer packets into discrete units called **Frames**.
- Adds header (source and destination MAC addresses) and trailer (error-checking CRC).

B. Physical Addressing (MAC Addressing)

- If frames are sent over a shared cable (LAN), the layer attaches a unique 48-bit **MAC Address** to identify the exact sender and receiver network interface card (NIC).

C. Flow Control

- Regulates transmission rate so that a fast sender does not swamp a slow receiver's internal memory buffer.

D. Error Control

- Adds error-detection codes (CRC, Checksum, Parity).
- Uses ARQ protocols (Stop-and-Wait, Go-Back-N, Selective Repeat) to retransmit corrupted or lost frames.

E. Access Control (MAC Sublayer)

- When multiple computers share a common wire or radio frequency, access control determines which computer is permitted to transmit at any given moment (CSMA/CD, Token Passing).

4. Real-Life Analogy

Think of sending an internal memo in a large office building:

- You don't just throw loose sheets of paper into the hallway (Raw bits).
- You place the memo inside an official company folder (**Framing**), write the recipient's room desk number (**MAC Address**), check that the person's inbox tray is not overflowing (**Flow Control**), and make sure no pages fell out (**Error Control**).

5. Dual Diagrams

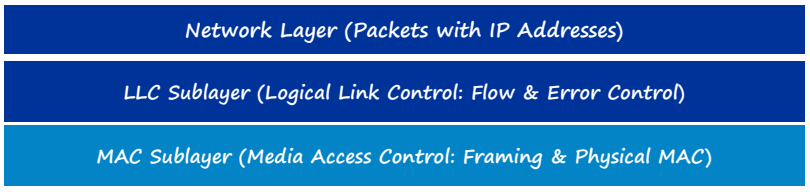


Diagram 1 (Standard Exam Diagram): Data Link Layer Architecture (LLC and MAC Sublayers)



Diagram 2 (Easy Understanding Diagram): Company Office Internal Memo Delivery Analogy

Diagram 1: Standard Exam Diagram

Shows the Data Link Layer divided into two sublayers: **LLC (Logical Link Control)** and **MAC (Media Access Control)** interfacing with Layer 3 and Layer 1.

Diagram 2: Easy Understanding Diagram

Shows a frame moving across a wire with Header, Payload Data, and Trailer securely attached.

6. Summary Box: Data Link Layer Services

Service Name	Purpose in Simple English	Mechanism Used
Framing	Adds start and end markers to group bits.	Byte Stuffing, Bit Stuffing.
Addressing	Identifies specific device on local cable.	48-bit MAC Address.
Flow Control	Prevents receiver buffer overflow.	Sliding Window (Stop-and-Wait, GBN).
Error Control	Catches noise flips & resends bad frames.	CRC, Checksum, Retransmission Timers.
Media Access	Prevents simultaneous colliding transmissions.	CSMA/CD, CSMA/CA, Token Passing.

7. Conclusion

By handling framing, addressing, speed adjustments, and noise recovery, the Data Link Layer provides an error-free, organized communication hop for the higher layers.

Q5. *(Paper-2)*

Explain GO-BACK-N sliding window protocol with diagram?

1. Introduction

Stop-and-Wait ARQ is terribly slow because the sender must wait for an ACK after sending every single frame. To keep high-speed fiber links busy, engineers developed **Pipelined Sliding Window Protocols**, the most popular of which is **Go-Back-N (GBN) ARQ**.

2. Definition

Go-Back-N ARQ is a pipelined sliding window protocol where the sender can transmit up to **N frames** without waiting for an acknowledgment. However, if any single frame is lost or damaged, the receiver silently discards all subsequent out-of-order frames, forcing the sender to **"Go Back N steps"** and retransmit all frames starting from the lost one.

3. Core Working Rules

A. Sender Window Size ($W_s = 2^m - 1$)

- For an m-bit sequence number, the sender window size is $N = 2^m - 1$.
- The sender can transmit all N frames inside the window before pausing.

B. Receiver Window Size ($W_r = 1$)

- The receiver accepts frames in **strict sequential order only**.
- If Frame 0, 1, 2, 3 are sent, and Frame 1 is lost:
- When Frame 2 arrives, the receiver discards it!
- When Frame 3 arrives, the receiver discards it!
- The receiver only wants Frame 1.

C. Cumulative Acknowledgments

- The receiver sends cumulative ACKs (e.g., ACK 3 means: *"I have received everything up to Frame 2; send Frame 3 next"*).

D. Timeout Retransmission

- The sender maintains a single timer for the oldest unacknowledged frame.
- If that timer expires, the sender retransmits **all unacknowledged frames** currently in the window.

4. Real-Life Analogy

Imagine an exam where a teacher grades 5 questions in strict order:

- If Question 2 is missing, the teacher stops grading and refuses to look at Questions 3, 4, and 5.
- The teacher hands the entire booklet back and says: *"Rewrite Question 2 and all questions after it!"*
- You must go back and rewrite Questions 2, 3, 4, and 5.

5. Dual Diagrams

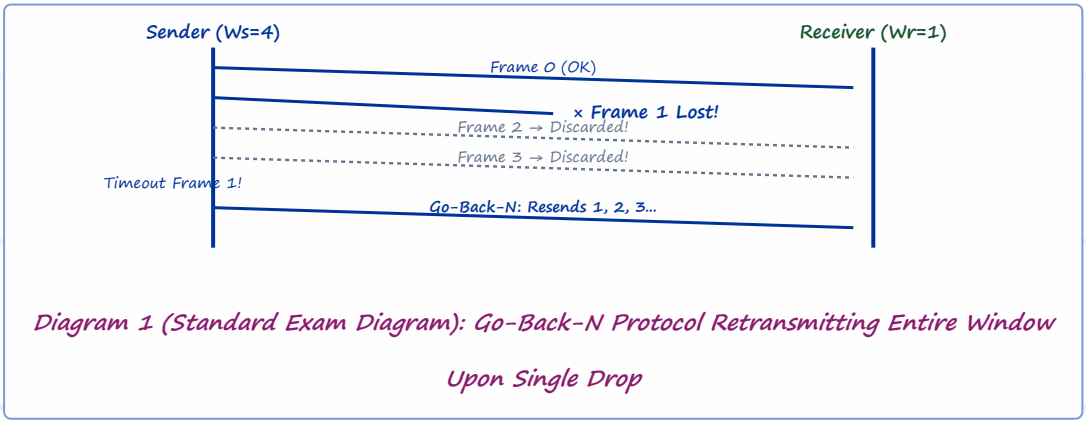


Diagram 1 (Standard Exam Diagram): Go-Back-N Protocol Retransmitting Entire Window Upon Single Drop

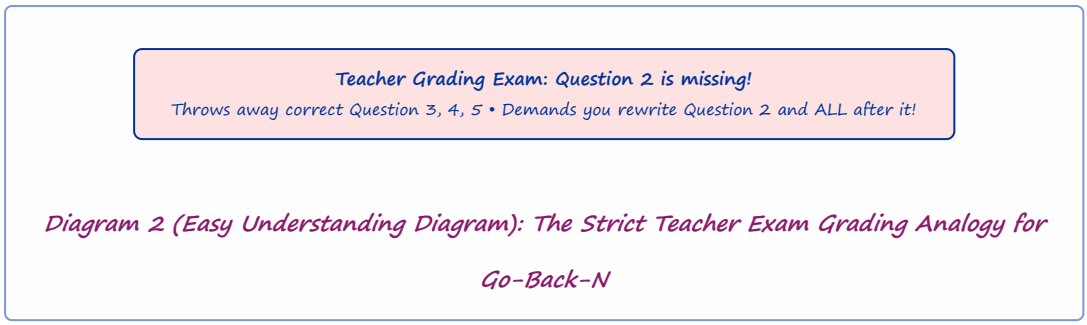


Diagram 2 (Easy Understanding Diagram): The Strict Teacher Exam Grading Analogy for Go-Back-N

Diagram 1: Standard Exam Diagram

Shows vertical timeline where Sender sends Frames 0, 1, 2, 3. Frame 1 gets lost. Frames 2 and 3 are discarded by receiver. Timer expires for Frame 1, and Sender goes back to retransmit Frames 1, 2, 3.

Diagram 2: Easy Understanding Diagram

Shows the physical sliding window rectangle enclosing frames [0, 1, 2, 3], sliding forward as ACKs arrive.

6. Advantages and Disadvantages of Go-Back-N

Advantages	Disadvantages
High Channel Utilization: Much faster than Stop-and-Wait because multiple frames are in flight.	**Wasted Retransmissions:** If one frame drops, perfectly good subsequent frames are thrown away.
Simple Receiver Buffer: Receiver only needs memory for **1 frame**, keeping hardware costs low.	**Sluggish on Noisy Channels:** On links with frequent packet drops, performance collapses due to constant re-sending.

7. Conclusion

Go-Back-N is an efficient, simple pipelined protocol that maximizes link speed when packet drop rates are low, requiring minimal memory at the receiver.

Q6. *(Paper-2)*

Mention different types of Error detection methods, Explain any two methods with example?

1. Introduction

Transmission media are never 100% free from electrical interference. Noise spikes can flip bits during transmission. To protect data integrity, network protocols use **Error Detection Methods**

that append redundant check bits to the data.

2. Four Primary Types of Error Detection Methods

- **Simple Parity Check (Single-bit Parity)**
- **Two-Dimensional (2D) Parity Check**
- **Checksum (One's Complement Addition)**
- **CRC (Cyclic Redundancy Check)**

3. Detailed Explanation of Method 1: Simple Parity Check

A. Working

- A single extra bit called the **Parity Bit** is added to the data block.
- **Even Parity:** The parity bit is chosen so that the total count of 1s in the block is **even**.
- **Odd Parity:** The parity bit is chosen so that the total count of 1s in the block is **odd**.

B. Example

- Suppose Data to send = `1001101` (Count of 1s = 4, which is already even).
- For Even Parity, Parity bit = `0`. Transmitted = `10011010`.
- If noise flips one bit during transit to `10111010` (Count of 1s becomes 5 — odd!), the receiver detects the error immediately and drops the frame.

C. Limitation

Cannot detect an **even number of errors** (if 2 bits flip simultaneously, the count of 1s remains even!).

4. Detailed Explanation of Method 2: Checksum (One's Complement)

A. Working

- The sender slices the data into equal k-bit words (usually 16 bits).
- All words are added together using one's complement addition (any carry bit from the top is added back to the bottom).
- The sender flips all bits of the sum (1's complement) to produce the **Checksum**.
- The checksum is sent along with the data words.
- The receiver adds all data words plus the checksum. If the result is all 1s (11111111), there is zero error!

B. Example

- Let data words be `1010` and `0111`.
- Add: `1010 + 0111 = 10001`. Carry 1 wrapped around: `0001 + 1 = 0010`.

- Invert bits: ext{Checksum} = 1101.
- Receiver adds: 1010 + 0111 + 1101 = 1111 (All 1s → Valid!).

5. Dual Diagrams

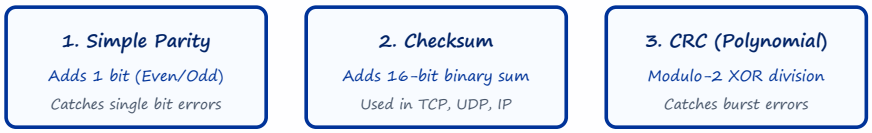


Diagram 1: Standard Exam Diagram – The Three Primary Error Detection Methods



Diagram 2: Easy Understanding Diagram – Bank Cashier Verification Analogy

Diagram 1: Standard Exam Diagram

Shows classification tree of the 4 Error Detection Methods with sub-branches.

Diagram 2: Easy Understanding Diagram

Shows data passing through a Parity Generator and Checksum Inverter before leaving the sender.

6. Summary Comparison Table of Error Detection Methods

Method	Error Detection Capability	Computational Complexity	Primary Usage
Simple Parity	Detects only single-bit errors; fails on 2-bit flips.	Minimal (simple XOR)	Asynchronous serial ports.
2D Parity	Detects burst errors up to a certain size.	Low	Legacy magnetic tape storage.
Checksum	Detects all odd-bit errors and most burst errors.	Moderate (software addition)	TCP, UDP, IPv4 Headers.
CRC	Detects 100% of single, double, and burst errors ≤ k.	High (hardware shift registers)	Ethernet, Wi-Fi, SATA disks.

7. Conclusion

While simple parity provides basic checks for low-speed links, Checksums and CRC provide robust mathematical error detection for modern high-speed internet and local area networks.

Q7. *(Paper-2)*

What is Burst Error? Find the Checksum for the following data: 10110011 10101011 01011010 11010101

1. Introduction & Definition of Burst Error

- **Burst Error:** A burst error means that **two or more consecutive bits** in a transmitted data block have been corrupted by a prolonged burst of electrical noise, lightning, or a physical loose connection.
- **Burst Length:** The distance from the first corrupted bit to the last corrupted bit (even if some bits inside the burst remain undamaged).
- *Example:* If sent data is `10000001` and received data is `10111101`, a 6-bit burst error occurred.

2. Step-by-Step Checksum Calculation

1. Given Data

Four 8-bit bytes of data:

- Byte 1 = `10110011`
- Byte 2 = `10101011`
- Byte 3 = `01011010`
- Byte 4 = `11010101`

2. What We Need to Find

- Sum of all four bytes using 8-bit One's Complement Arithmetic.
- The final 8-bit **Checksum** to be transmitted.
- Verification at receiver.

3. Step-by-Step Addition

- **Step 1: Add Byte 1 and Byte 2:**

```
10110011 (Byte 1)
+ 10101011 (Byte 2)
----------
101011110 (9 bits: 1 carry bit + 8 sum bits)
```

Wrap around the carry bit:

01011110 + 1 = 01011111

- **Step 2: Add Byte 3 to the result:**

```
01011111 (Current Sum)
+ 01011010 (Byte 3)
----------
10111001 (8 bits, no carry)
```

- **Step 3: Add Byte 4 to the result:**

```
10111001 (Current Sum)
+ 11010101 (Byte 4)
----------
110001110 (9 bits: 1 carry bit + 8 sum bits)
```

Wrap around the carry bit:

10001110 + 1 = 10001111

4. Calculate the Checksum (1's Complement)

Flip every bit of the final sum (10001111):

ext{Checksum} = \sim(10001111) = \mathbf{01110000}

5. Final Boxed Answer

Final Sum = 10001111 **Transmitted Checksum = 01110000**

6. Receiver Verification

The receiver adds all 4 data bytes plus the received checksum:

ext{Sum} = 10001111 + 01110000 = \mathbf{11111111}

Inverting all bits yields **00000000 (Zero)**. The receiver concludes that the data is 100% correct and accepts the frame.

3. Real-Life Analogy

A checksum is like **Weighing a Postal Carton at the Post Office**:

- You put 4 packages weighing 5 kg, 3 kg, 2 kg, and 4 kg into a box (Total = 14 kg).
- You stamp "14 kg" on the lid.
- The destination clerk weighs the box. If it weighs exactly 14 kg, nothing was lost or stolen.

4. Dual Diagrams

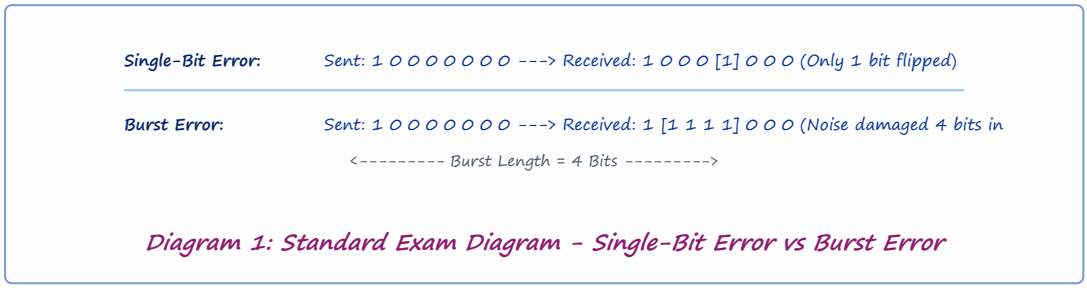


Diagram 1: Standard Exam Diagram – Single-Bit Error vs Burst Error

Diagram 2: Easy Understanding Diagram – CD Scratch Burst Error Analogy

Diagram 1: Standard Exam Diagram

Shows the step-by-step column addition of the four 8-bit bytes, carry wraparound, and 1's complement inversion.

Diagram 2: Easy Understanding Diagram

Shows a Burst Error waveform where a lightning spark corrupts a cluster of 5 adjacent bits.

5. Conclusion

Checksum provides a fast, lightweight method to verify entire multi-byte packets in software, which is why it is used across the TCP/IP suite.

Q8. *(Paper-2)*

Explain Selective Repeat sliding window protocol with diagram?

1. Introduction

While Go-Back-N (GBN) improves throughput over Stop-and-Wait, it has a serious flaw: if one frame drops, the receiver throws away all subsequent frames, forcing the sender to retransmit everything. **Selective Repeat ARQ** fixes this inefficiency by retransmitting **only the specific frame that was lost**.

2. Definition

Selective Repeat ARQ is an advanced sliding window protocol where the receiver possesses a dedicated buffer of size $W_r > 1$, enabling it to accept and store out-of-order frames. The sender retransmits *only* the corrupted or lost frames.

3. Key Differences from Go-Back-N

- **Sender Window Size (W_s):** Must be equal to half the sequence number space:

$W_s = 2^{m - 1}$

- **Receiver Window Size (W_r):** Equal to sender window size ($W_r = W_s = 2^{m-1}$).
- **Negative Acknowledgment (NAK):** The receiver can explicitly send a `NAK 2` telling the sender: *"Frame 2 was corrupted; resend Frame 2 only!"*
- **Individual Timers:** The sender maintains a separate retransmission timer for every single frame sent.

4. Operational Walkthrough

- Sender transmits Frames 0, 1, 2, 3.
- Frame 1 is lost on the wire.
- Frames 2 and 3 arrive safely. The receiver **does not throw them away**; it stores them in its buffer!
- Receiver sends a **NAK 1** (or timer expires for Frame 1).
- Sender retransmits **only Frame 1**.
- When Frame 1 arrives, the receiver glues Frames 0, 1, 2, 3 together in correct order and delivers them to the Network Layer.

5. Real-Life Analogy

Imagine ordering a 5-volume encyclopedia set:

- Books 1, 3, 4, 5 arrive, but Book 2 was damaged in transit.
- In Go-Back-N, you throw Books 3, 4, and 5 into the trash and demand all 4 books be resent.
- In Selective Repeat, you keep Books 3, 4, and 5 on your bookshelf, call the bookstore, and ask them to resend **only Book 2**.

6. Dual Diagrams

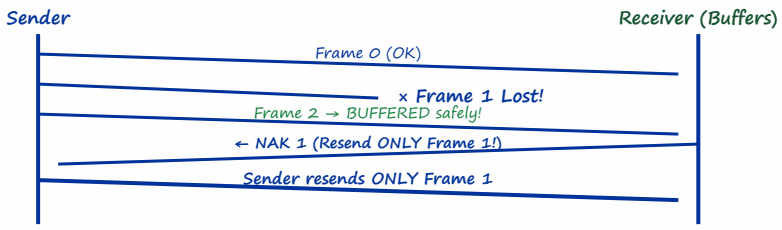


Diagram 1 (Standard Exam Diagram): Selective Repeat Buffering Out-of-Order Frames and Retransmitting Only Lost Frame



Diagram 2 (Easy Understanding Diagram): Missing Book Replacement Analogy for Selective Repeat

Diagram 1: Standard Exam Diagram

Shows vertical timeline: Sender transmits 0, 1, 2. Frame 1 is lost. Receiver buffers Frame 2 and sends NAK 1. Sender retransmits only Frame 1. Receiver delivers 0, 1, 2 in order.

Diagram 2: Easy Understanding Diagram

Shows receiver sliding window buffer holding an empty slot for missing Frame 1 while safely buffering Frame 2.

7. Box Comparison: Go-Back-N vs Selective Repeat

Feature	Go-Back-N ARQ	Selective Repeat ARQ
Receiver Window Size	$W_r = 1$ (Cannot buffer out-of-order frames).	$W_r = 2^{m-1}$ (Buffers out-of-order frames).
Retransmissions	Retransmits lost frame **plus all following frames**.	Retransmits **only the specific lost frame**.
Bandwidth Efficiency	Low on noisy channels (wastes bandwidth).	**Extremely High** (minimal retransmissions).
Sender/Receiver Memory	Simple; needs minimal buffer.	Complex; requires individual timers and sorting buffers.
Window Size Formula	$W_s = 2^m - 1$	$W_s = 2^{m-1}$

8. Conclusion

Selective Repeat maximizes bandwidth efficiency by eliminating duplicate retransmissions, making it the preferred protocol for high-performance networks despite its extra memory requirements.

Q9. *(Paper-3)*

Explain fixed-size framing with an example.

1. Introduction

At the Data Link Layer, raw bits from the physical layer must be packaged into bounded chunks called **Frames**. The simplest way to do this is **Fixed-Size Framing**.

2. Definition

Fixed-Size Framing is a framing method where every single transmitted frame has the exact same predetermined length (fixed number of bytes). There is no need for frame delimiter flags because the receiver simply counts the bits.

3. Working Mechanism

- The network standard dictates the fixed frame length (e.g., exactly 1000 bits or 53 bytes).
- The receiver counts bits as they arrive:
- When bit count reaches 53 bytes → Frame 1 is complete!
- The next 53 bytes → Frame 2!
- If a message is smaller than the fixed size, **Padding Bits (zeros)** are added to fill the space.

4. Real-World Example: ATM (Asynchronous Transfer Mode) Networks

- ATM technology uses fixed-size frames called **Cells**.
- Every ATM cell is strictly **53 Bytes long** (5 Bytes of Header + 48 Bytes of Payload).
- Because the size never changes, hardware switches can route cells at blistering speeds without inspecting packet boundaries.

5. Real-Life Analogy

Think of an **Egg Carton with 12 Slots**:

- Every egg carton holds exactly 12 eggs.

- The packaging machine does not need a sensor to measure the carton size — it just counts to 12 eggs, seals the box, and starts the next box.
- If you only have 10 eggs, you put in 2 dummy plastic eggs (Padding) to keep the carton full.

6. Dual Diagrams



Diagram 1: Standard Exam Diagram – Fixed-Size Framing (ATM 53-Byte Cell)



Diagram 2: Easy Understanding Diagram – Fixed-Size Egg Carton Analogy

Diagram 1: Standard Exam Diagram

Shows continuous incoming bit stream divided into equal fixed blocks of length L.

Diagram 2: Easy Understanding Diagram

Shows 53-byte ATM cell structure (5-byte Header + 48-byte Data Payload).

7. Advantages and Disadvantages of Fixed-Size Framing

Advantages	Disadvantages
No Delimiter Flags Needed: Simple bit counting identifies frame boundaries.	**Wasted Bandwidth (Padding):** If data is tiny, padding bits waste cable capacity.
Ultra-Fast Hardware Routing: Switches process fixed cells predictably.	**Rigid:** Cannot adapt to bursty, variable-sized Internet traffic.

8. Conclusion

Fixed-size framing offers hardware simplicity and predictable switching delays, but has largely been superseded by variable-size framing in modern IP networks to avoid padding waste.

Q10. *(Paper-3)*

Explain Checksum with suitable example.

1. Introduction

Error detection is critical to ensure that data received over a network matches what was sent. One of the most widely used methods across Internet protocols (TCP, UDP, IP) is the **Checksum**.

2. Definition

A **Checksum** is an error-detection code generated by summing the binary values of a sequence of data words using one's complement addition, and appending the one's complement of that sum to the message.

3. Step-by-Step Numerical Example

A. Sender Side Calculation

Suppose we want to send two 4-bit numbers:

- Word 1 = 0111 (7 in decimal)
- Word 2 = 1011 (11 in decimal)

• **Add Word 1 and Word 2:**

$$0111 + 1011 = 10010 \text{ (5 bits)}$$

• **Handle the Carry Bit (Wraparound):**

$$0010 + 1 = \mathbf{0011} \text{ (Sum)}$$

• **Take One's Complement (Flip Bits):**

$$\text{Checksum} = \sim(0011) = \mathbf{1100}$$

• **Transmit:** Word 1 (0111), Word 2 (1011), and Checksum (1100).

B. Receiver Side Verification

The receiver receives all three 4-bit blocks and adds them together:

• $0111 + 1011 = 10010 \Rightarrow 0010 + 1 = 0011$

• Add Checksum:

$$0011 + 1100 = \mathbf{1111}$$

• Take One's Complement:

$$\sim(1111) = \mathbf{0000} \text{ (Zero Error!)}$$

The receiver accepts the data as 100% valid.

4. Real-Life Analogy

A checksum is like a **Restaurant Cash Register Bill**:

- Burger = \7
- Shake = \11
- Total = \18
- If you hand the cashier a \20 note and the change doesn't add up correctly, you know an accounting mistake occurred.

5. Dual Diagrams

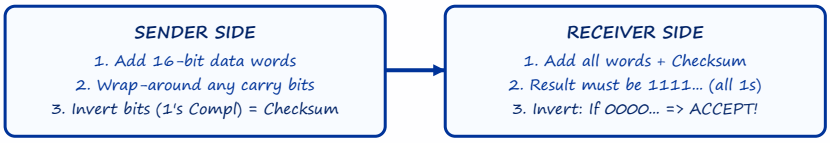


Diagram 1: Standard Exam Diagram – Checksum Sender and Receiver Logic Flow



Diagram 2: Easy Understanding Diagram – Supermarket Receipt Total Analogy

Diagram 1: Standard Exam Diagram

Shows Sender Checksum Generator block adding data words, inverting sum, and Receiver Checksum Checker verifying if result is all 1s.

Diagram 2: Easy Understanding Diagram

Shows 4-bit addition with carry bit circling around to the bottom.

6. Advantages and Disadvantages of Checksum

Advantages	Disadvantages
Very Easy in Software: Simple integer addition is fast for CPUs.	Does not catch errors where bits swap values without changing total sum.
Very Few Extra Bits: Only adds 16 bits for an entire packet.	Weaker error detection than polynomial CRC.

7. Conclusion

Checksum is a fast and simple method that computers use to catch errors in packet headers across the Internet.

Q11. *(Paper-3)*

Explain one-bit sliding window protocol with diagram and example.

1. Introduction

The **One-Bit Sliding Window Protocol** is the fundamental building block of all bidirectional sliding window protocols. It achieves full-duplex communication where data and acknowledgments travel in both directions simultaneously using a concept called **Piggybacking**.

2. Definition

A **One-Bit Sliding Window Protocol** is a flow and error control protocol where both sender and receiver maintain a window size of W = 1, using a 1-bit sequence number (alternating between 0 and 1).

3. Core Working Features

- **Sequence Numbers:** Uses only 0 and 1.
- **Piggybacking:** When Station A sends data to Station B, it attaches the ACK for Station B's previous message inside the same frame header, saving bandwidth.
- **Stop-and-Wait Nature:** Because window size is 1, a station cannot send a new data frame until its previous frame is acknowledged.

4. Operational Example

• Station A sends Frame with ext{Seq}=0, ext{Ack}=0.

• Station B receives it, processes data, and sends back its own data with ext{Seq}=0, acknowledging A's frame with ext{Ack}=1 (piggybacking!).

• Station A receives the ACK and sends its next Frame with ext{Seq}=1, ext{Ack}=1.

5. Dual Diagrams

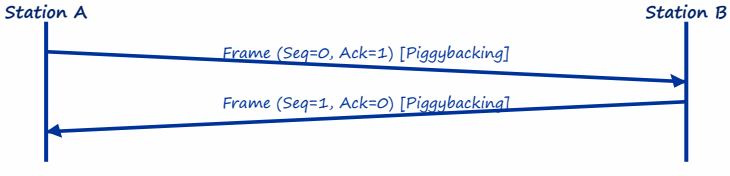


Diagram 1: Standard Exam Diagram - One-Bit Sliding Window Piggybacking Flow



Diagram 2: Easy Understanding Diagram – Two-Way Conversation Analogy

Diagram 1: Standard Exam Diagram

Shows full-duplex timeline between Host A and Host B showing bidirectional frames with piggybacked Seq and Ack numbers.

Diagram 2: Easy Understanding Diagram

Shows Piggybacking: Passenger Data + Conductor ACK Ticket traveling inside the same bus.

6. Advantages and Disadvantages

Advantages	Disadvantages
Piggybacking Saves Bandwidth: Combines data and ACKs into a single frame.	**Low Link Utilization:** Window size of 1 leaves high-speed links largely idle.
Full Duplex: Both stations can transmit simultaneously.	If a station has no reverse data to send, ACK delivery is delayed.

7. Conclusion

The One-Bit Sliding Window protocol bridges simple Stop-and-Wait with full-duplex communications, establishing the foundation for modern protocols like TCP.

Q12. *(Paper-3)*

Explain Point-to-Point Protocol (PPP).

1. Introduction

When an individual connects their home computer or router to an Internet Service Provider (ISP) over a dedicated telephone or optical fiber line, they use the **Point-to-Point Protocol (PPP)**.

2. Definition

PPP (Point-to-Point Protocol) is a Data Link Layer protocol designed to establish a direct connection between two networking nodes, providing authentication, link configuration, and network-layer protocol multiplexing.

3. Three Core Components of PPP

- **Framing Method:** An HDLC-like frame structure with boundary flags (`01111110`) and byte-stuffing for transparency.
- **LCP (Link Control Protocol):** Establishes, tests, configures, and cleanly terminates the data link connection.
- **NCP (Network Control Protocols):** A series of helper protocols that configure different network-layer protocols (such as IPCP for IPv4/IPv6).

4. The 6-Phase Life Cycle of a PPP Connection

- **Phase 1: Dead:** Link is physically inactive.
- **Phase 2: Establish:** LCP exchanges configuration packets.
- **Phase 3: Authenticate:** Verifies user identity using **PAP** (Password Authentication Protocol) or **CHAP** (Challenge Handshake Authentication Protocol).
- **Phase 4: Network:** NCP assigns an IP address to the client.
- **Phase 5: Open:** User data is actively transferred.
- **Phase 6: Terminate:** LCP closes the connection when the user logs off.

5. Real-Life Analogy

PPP is like **Calling a Bank Hotline**:

- You dial the number (**Establish Link**).
- The agent asks for your name and security PIN (**Authentication via PAP/CHAP**).
- The agent pulls up your account (**Network Configuration**).
- You complete your banking transactions (**Data Transfer**).
- You say goodbye and hang up the phone (**Termination**).

6. Dual Diagrams



Diagram 1 (Standard Exam Diagram): Point-to-Point Protocol (PPP) Frame Architecture

PPP Life Cycle: Calling Bank Phone Analogy

Dead → Establish (LCP) → Authenticate (PAP/CHAP) → Network (IPCP) → Open → Terminate

Diagram 2 (Easy Understanding Diagram): PPP Connection Lifecycle Sequence

Diagram 1: Standard Exam Diagram

Shows the PPP Frame Format: Flag (01111110) • Address (11111111) • Control (00000011) • Protocol (2 bytes) • Payload Data • FCS Checksum • Flag.

Diagram 2: Easy Understanding Diagram

Shows the PPP State Transition Diagram (Dead → Establish → Authenticate → Network → Open → Terminate).

7. Conclusion

PPP is the universal protocol for point-to-point connections, combining authentication, link testing, and multi-protocol flexibility into a standardized frame.

Q13. *(MID-1)*

Explain fixed-size and variable-size framing techniques with suitable examples.

1. Introduction

At the Data Link Layer, data must be packaged into frames. Depending on how frame boundaries are marked, framing is classified into **Fixed-Size Framing** and **Variable-Size Framing**.

2. Fixed-Size Framing

- **How It Works:** Every frame has the exact same predefined length. The receiver simply counts bits.
- **Example: ATM Networks** use fixed 53-byte cells.
- **Pros/Cons:** Fast hardware processing, but wastes bandwidth due to padding when messages are small.

3. Variable-Size Framing

In modern networks, data messages vary in length (from 64 bytes to 1500 bytes in Ethernet). To mark boundaries, two techniques are used:

A. Character-Oriented Framing (Byte Stuffing)

- Special delimiter flag bytes like FLAG (01111110) mark the start and end of a frame.
- **Byte Stuffing:** If the data payload naturally contains the word FLAG, the sender inserts an Escape byte (ESC) before it: ESC FLAG. The receiver strips ESC and treats FLAG as pure data.

B. Bit-Oriented Framing (Bit Stuffing)

- Uses the flag pattern 01111110 (six consecutive 1s).
- **Bit Stuffing Rule:** Whenever the sender sees **five consecutive 1s** (11111) in the data, it automatically injects a **0** after them: 111110.
- The receiver watches for five 1s: if the next bit is 0, it strips the 0. If the next bit is 1, it checks for the flag pattern!

4. Dual Diagrams

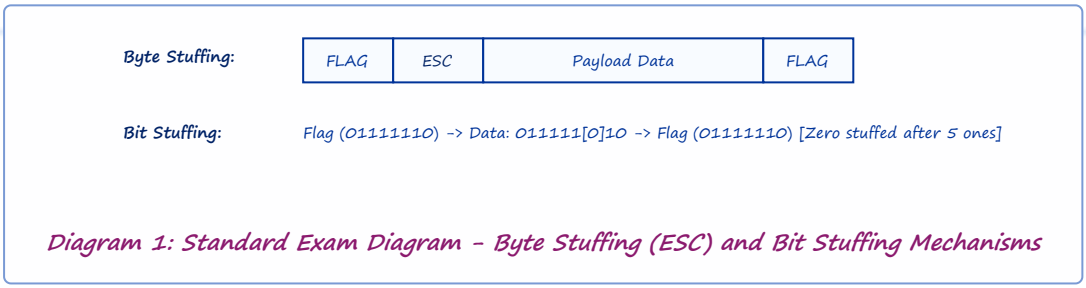


Diagram 1: Standard Exam Diagram - Byte Stuffing (ESC) and Bit Stuffing Mechanisms



Diagram 2: Easy Understanding Diagram - Envelope Boundaries Analogy

Diagram 1: Standard Exam Diagram

Shows Byte Stuffing (ESC insertion) and Bit Stuffing (inserting 0 after five 1s: 0111110 becomes 01111100).

Diagram 2: Easy Understanding Diagram

Shows an envelope with official green wax seals at the top and bottom to prove nobody tampered with the letter boundaries.

5. Box Comparison Table: Fixed-Size vs Variable-Size Framing

Feature	Fixed-Size Framing	Variable-Size Framing
Frame Length	Constant across all frames (e.g., 53 Bytes).	Varies dynamically based on data size (64 to 1500 B).
Boundary Markers	None needed; uses bit counters.	Uses Delimiter Flags, Byte Stuffing, or Bit Stuffing.
Padding Requirement	High (adds dummy zeros to small packets).	Minimal to None.
Bandwidth Efficiency	Lower due to padding.	**Much Higher**; transmits only needed bytes.
Modern Examples	ATM, Sonet.	Ethernet, Wi-Fi, HDLC, PPP.

6. Conclusion

Fixed-size framing offers hardware simplicity for specialized networks, while variable-size framing with bit stuffing is the standard for the modern Internet because it eliminates padding waste.

Q14. *(MID-1)*

Explain Sliding Window protocols with reference to One-bit, Go-Back-N and Selective Repeat protocols.

1. Introduction

Sliding window protocols are the gold standard for reliable, full-duplex data transmission in computer networks. They allow multiple frames to be in flight simultaneously, maximizing the use of high-speed transmission links.

2. Detailed Comparison of the Three Sliding Window Protocols

A. One-Bit Sliding Window Protocol

- **Window Sizes:** Sender Window $W_s = 1$, Receiver Window $W_r = 1$.
- **Working:** Alternates sequence numbers between 0 and 1. Transmits one frame and waits for an ACK before sending the next.
- **Efficiency:** Very low; operates like Stop-and-Wait with piggybacking.

B. Go-Back-N (GBN) Protocol

- **Window Sizes:** $W_s = 2^m - 1$, $W_r = 1$.

- **Working:** Sender transmits up to N frames continuously. If Frame k drops, receiver throws away all subsequent frames. Sender must **go back** and retransmit Frame k and all frames after it.
- **Efficiency:** High on clean lines, but poor on noisy links due to duplicate retransmissions.

C. Selective Repeat (SR) Protocol

- **Window Sizes:** $W_s = 2^{m-1}$, $W_r = 2^{m-1}$.
- **Working:** Receiver has a large buffer to store out-of-order frames. If Frame k drops, receiver holds the other frames and asks for **only Frame k** to be resent.
- **Efficiency:** **Highest** of all protocols; zero wasted bandwidth.

3. Dual Diagrams

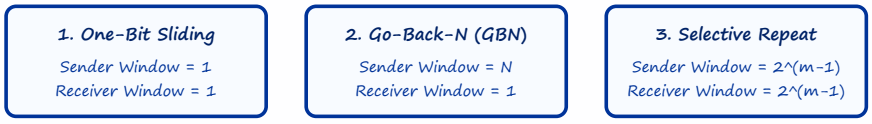


Diagram 1: Standard Exam Diagram – Window Sizes of the Three Sliding Window Protocols



Diagram 2: Easy Understanding Diagram – Replacing Broken Items Analogy

Diagram 1: Standard Exam Diagram

Shows side-by-side comparison of sliding windows for One-Bit ($W=1$), Go-Back-N ($W_s=N$, $W_r=1$), and Selective Repeat ($W_s=W_r=N/2$).

Diagram 2: Easy Understanding Diagram

Illustrates how Go-Back-N throws away good packets upon 1 error, whereas Selective Repeat buffers good packets and fills the single missing hole.

4. Master Comparison Box of Sliding Window Protocols

Parameter	One-Bit Protocol	Go-Back-N (GBN)	Selective Repeat (SR)
Sender Window (W_s)	1	$2^m - 1$	2^{m-1}

Parameter	One-Bit Protocol	Go-Back-N (GBN)	Selective Repeat (SR)
Receiver Window (W_r)	1	1	2^{m-1}
Out-of-Order Buffering	No	No (Discarded)	**Yes (Stored in buffer)**
Retransmissions	Single frame	**Lost frame + all subsequent**	**Only the lost frame**
Acknowledgment Type	Individual	**Cumulative**	**Individual / Negative (NAK)**
Hardware Complexity	Minimal	Low (simple receiver)	High (sorting buffers & timers)
Bandwidth Efficiency	Lowest	Medium	**Highest**

5. Conclusion

One-Bit is the simplest bidirectional protocol; Go-Back-N provides high throughput with minimal receiver memory; Selective Repeat delivers maximum channel efficiency by eliminating redundant retransmissions.
